Gagan Kumar Rajulapati

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Education

2026/05 Hyderabad, India
Data Science
Innomatics Research Science
Coursework in Python, Statistics, Data Analysis, SQL + PowerBI, ML, DL, GenAI, NLP, Computer Vision

2025/05 Krishnankoil, India
Bachelor of Computer Science – CSE(AIML)
kalasalingam University

Skills

PYTHON
- Python: Anaconda, Jupyter Notebook, Colab Notebook, syntax, Identifiers & Operators, Array, Numpy, Data Manipulation, Pandas

SQL + Power BI
- MYSQL, NoSQL, CRUD operation, RDBMS, Data Exploration and Data Filtering, DQL and Operators, Clauses, Joins, ACID, COMMIT, ROLLBACK.

Deep Learning
- AI, MLAND DL, Biological Neuron, Google Colab, Tensorflow, Neural Network, Forward & Backward Propagation, Optimizers, MLP

CNN & Computer Vision
- Open CV, Convolutional, TransferLearing, Object Detection, YOLO.

GEN AI
- Large Language Models, OpenAI, DALL-E, GPT-3/GPT-4, Gemini, LLaMA, LangChain

Advanced Statistics
- Descriptive Statistics, Probability Distribution, Data Gathering Techniques, Inferential Statistics.

Power BI
- Power BI workflow, Visualisation, Trend Data viz, Power Queries, Power Pivot, DAX, DA Expression, Web & RLS, Visual Interactions, Drill Through.

Machine Learning
- Supervised, Unsupervised, ML Algorithms, Validation Methods, Naive Bayes, Linear Algebra, K-NN, Hyperparameters, Dimensionality, Decision Tree, Linear regression, Evaluation Metrics, Regression Techniques, Q-Q Plot, MSE & RMSE, Polynomial regression, Regularization Techniques, Logistic Regression, Hierarchical Clustering.

Natural Language processing
- NLTK, spacy, BOW, TF-IDF, Glove, Fast text, RNNs, LSTMs, Auto Encoders,BERT

Projects

1. **Heart & Lung Sound Classification System - ML**
- Developed a machine learning system to classify heart and lung sound recordings from audio files using Python.
- Extracted MFCC audio features using Librosa to perform audio signal processing and feature engineering.
- Trained a Random Forest classifier using Scikit-learn to predict heart, lung, or mixed sound categories.
- Built an interactive Streamlit web application with audio upload, waveform visualization, and Mel spectrogram analysis
- Generated Implemented prediction confidence scoring to improve model interpretability and user understanding.

2. **CareerPilot AI – Resume Analysis & Smart Job Recommendation System**
- Developed an AI-based resume analyzer using NLP to extract skills and recommend relevant job roles.
- Built a resume–job matching system using semantic similarity to calculate compatibility scores.
- Implemented FastAPI backend and Streamlit frontend for resume upload, skill extraction, and job recommendations.
- Designed a modular AI agent pipeline integrating resume parsing, job scraping, and automated application simulation..

3. **Cyber Fraud Patterns & Insights Using** Python
- large-scale datasets using Python (Pandas, NumPy) to uncover trends, patterns, and data inconsistencies.
- Cleaned and preprocessed 10,000+ records, handling missing values, duplicates, and outliers to ensure data quality.
- Performed Univariate, Bivariate, and Multivariate analysis to study relationships between key variables.
- Built 20+ visualizations using Matplotlib and Seaborn to clearly communicate insights to stakeholders
- Generated actionable insights that improved data-driven decision-making and analytical accuracy

4. **Autonomous AI Financial Crime Intelligence System – AI/ML Fraud Detection Platform**
- Developed an AI-powered financial crime detection system to analyze 6M+ banking transactions using Python and ml
- Implemented Isolation Forest anomaly detection to identify suspicious transactions and high-risk accounts.
- Built an interactive Streamlit dashboard with Plotly visualizations for fraud trends, transaction analysis, and investigation tools.
- Performed fraud network analysis and money-flow visualization using NetworkX, PyVis, and Sankey diagrams.

5. **Employee Management System – SQL Database & Analytics**
- Designed a relational SQL database with 6 interconnected tables supporting 5,000+ employee, salary, leave & qualification records.
- Executed 30+ analytical SQL queries to analyze salary distribution, payroll variance, leave activity, job roles & qualification levels.
- Identified top-paid employees and 10+ departments with the highest salary allocation using advanced joins, aggregations & window functions.
- Assessed 4 years of leave data, revealing correlations that helped HR reduce manual analysis time by 30%.
- Optimized data structure with proper key Optimized database structure with proper keys, indexes, joins & constraints, improving query performance.

Certificates

- Certification in data science Innomatics Research labs
- Exploratory Data Analysis from Innomatics Research Labs
- Certification in data analysis Innomatics Research lab.

Languages

- English
- Hindi
- Telugu
- French
- Tamil